Tahmid Hasan Sakib

Cookeville, Tennessee | tsakib42@tntech.edu | LinkedIn | GitHub | Google Scholar

PROFESSIONAL SUMMARY

M.S. researcher working across embedded interface testing, bench debugging, Python test automation, and academic digital design. Physical Jetson testbeds and AVR/Arduino prototyping connect hardware behavior with software validation.

TECHNICAL SKILLS

RTL and digital design: Verilog (introductory), Icarus Verilog, GTKWave, Cadence digital VLSI coursework; pipeline, ALU, register-file, and hazard/forwarding concepts.

Embedded test and simulation: Python, C/C++, Linux, Git, UART/I2C/SPI/GPIO/PWM/Ethernet, Jetson TX2/AGX Orin, AVR/Arduino, MATLAB, LTspice, Cisco Packet Tracer.

HARDWARE AND RESEARCH EXPERIENCE

Hardware Engineer Intern | Robi Axiata Limited, Dhaka, Bangladesh | December 2019-January 2020

- Prototyped and tested AVR/Arduino systems for embedded hardware/software validation.
- Simulated and documented hardware and communication behavior with MATLAB, LTspice, and Cisco Packet Tracer.

Graduate Research Assistant | Tennessee Technological University, Cookeville, TN | May 2024-Present

- Brought up selected interfaces on Jetson AGX Orin and ran bench-level checks and Python-based test automation.
- Validated QCar1 Jetson TX2 camera/LiDAR sensing against clean and printed-sign physical runs, recording configuration differences.
- Used clean-control baselines and log replay to investigate unexpected perception/control behavior.
- Evaluated study-specific QCar2 AGX Orin sensing and secure ROS 2 configurations separately from the non-ROS QCar1 platform.

SELECTED HARDWARE AND DIGITAL DESIGN PROJECTS

Five-Stage Pipelined RISC Processor | Verilog, Icarus Verilog, GTKWave | 2026

Implemented and simulated a 32-bit IF/ID/EX/MEM/WB processor with ALU, register file, memories, pipeline registers, hazard detection, and forwarding; used arithmetic, load/store, and branch testbenches.

Digital VLSI and Microprocessor Systems | Tennessee Tech coursework

Completed Introduction to Digital VLSI (Fall 2024) and Microprocessor Systems (Fall 2025); practiced Cadence digital simulation and microprocessor interfacing labs.

QCar physical testbed validation | Jetson TX2 and AGX Orin

QCar1 camera/LiDAR route and separate QCar2 secure ROS 2/perception studies supported physical sensor-system evaluation.

EDUCATION

Tennessee Technological University | M.S., Electrical and Computer Engineering, expected December 2026 | GPA 4.00/4.00

Master's thesis defended; official degree conferral pending.

Military Institute of Science and Technology, Dhaka, Bangladesh | B.Sc., Computer Science and Engineering, August 2021

SELECTED PUBLICATION

First author, physical QCar1 adversarial-perception study, IEEE Access, accepted for publication, 2026.